Question 3 — Design, Covariance and Gram Matrices:

Cleaning, Principal Component Analysis, and Clustering

SIMC 2026 Endeavour Challenge (Parts a–e)

Abstract

We analyse a design matrix of $N = 1000$ infrared spectra at $M = 3401$ frequencies. Thresholding the row-norm distribution at its spectral gaps removes 40 empty vials and 36 high-norm outliers, leaving $\widetilde{N} = 924$ spectra. The covariance matrix $C = \frac{1}{\widetilde{N}} \widetilde{X}^\top \widetilde{X}$ has total variance $\operatorname{tr} C \approx 7.90$, dominance ratio $\lambda_1/\lambda_2 \approx 3.53$, and a sharp elbow at $k^\star = 8$ signal eigenvalues; the "95% of variance" rule is shown to be misleading (it needs 667 components). The leading eigenvector is an interpretable difference-spectrum, and projection onto the top four eigenvectors resolves the data into ≈ 25 chemical concoctions.

1 Setup

$X \in \mathbb{R}^{N \times M}$ has rows = *samples* (vials) and columns = *features* (frequencies); X_{ij} is the absorbance of sample i at frequency j. The supplied array has shape 3401×1000: it is stored *transposed*, and its feature axis is 3401, not the 3041 quoted in the prompt (a digit transposition). We transpose so rows index samples and use $M = 3401$. The *Euclidean norm* $\|x\| = (\sum_j x_j^2)^{1/2}$ summarises a spectrum's overall magnitude.

2 Part (a): cleaning

Empty vials record only background noise: small, flat, hence small norm. Sorting the 1000 norms reveals a clean gap — 40 vials at $\|x\| \approx 1.6$, then a void until the bulk at $\|x\| \approx 4.3$ — so we threshold at the gap midpoint (a robust, constant-free choice). This flags exactly $T = 40$ empties (rows 960–999). The upper tail holds a *second* error: 36 vials at $\|x\| \approx 8.9$ (rows 924–959), detached by a gap $\approx 5\times$ wider than any internal gap. We retain i iff $\tau_{\text{lo}} \le \|x_i\| \le \tau_{\text{hi}}$ via a Boolean keep-mask (non-destructive, composable, index-preserving), leaving $\widetilde{N} = 924$ spectra. Validation: removed spectra are flat while retained ones are peaked, and empties are 50.2% exact zeros versus 24.5% (background noise clipped at zero is exactly 0 with probability $\approx \frac{1}{2}$). Section 6 shows the 36 outliers form one isolated cluster whose removal drops the cluster count by exactly one, so the decision is clean (and reversible).

3 Part (b): variance and the covariance matrix

Mean-centering, $\widetilde{X}_{ij} = X_{ij} - \bar{X}_j$ with $\bar{X}_j = \frac{1}{\widetilde{N}} \sum_i X_{ij}$, removes the shared mean spectrum so the analysis describes how samples *differ*. The covariance matrix is $C = \frac{1}{\widetilde{N}} \widetilde{X}^\top \widetilde{X} \in \mathbb{R}^{M \times M}$.

Proposition 1. *C is symmetric and $C_{jj} = \frac{1}{\widetilde{N}} \sum_k \widetilde{X}_{kj}^2 = \mathrm{Var}_j$, the variance of feature j.*

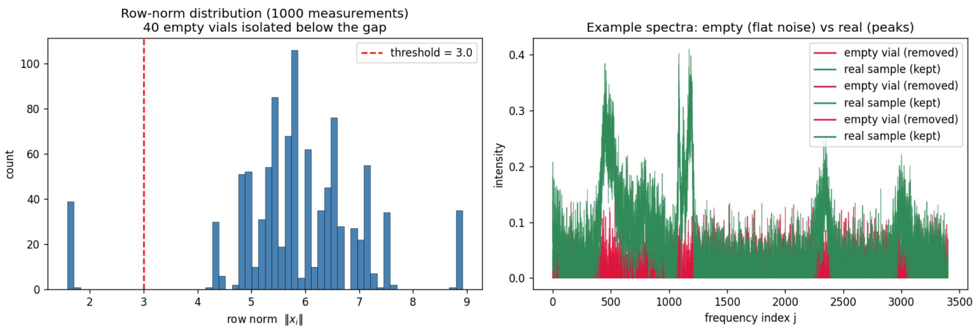

Figure 1: Row-norm distribution: 40 empty vials below the gap; example empty (flat) vs genuine (peaked) spectra.

Proof. $C^\top = \frac{1}{N}(\widetilde{X}^\top \widetilde{X})^\top = C$; setting $i = j$ and using zero column-means gives the variance. $\square$

Proposition 2. $\operatorname{tr} C = \sum_j \operatorname{Var}_j$ *(total variance) is invariant under orthogonal change of basis, and equals* $\sum_i \lambda_i$*.*

Proof. $\operatorname{tr}(Q^\top C Q) = \operatorname{tr}(C Q Q^\top) = \operatorname{tr} C$ for $Q^\top Q = I$; diagonalising $C = Q \Lambda Q^\top$ gives $\operatorname{tr} C = \sum_i \lambda_i$. $\square$

Empirically $\operatorname{tr} C \approx 7.90$. Variance concentrates in a few narrow molecular bands (Figure 2); the maximum is at $j^\star = 910$ with $C_{j^\star j^\star} \approx 0.0199$. Most features are flat, so the data is effectively low-dimensional.

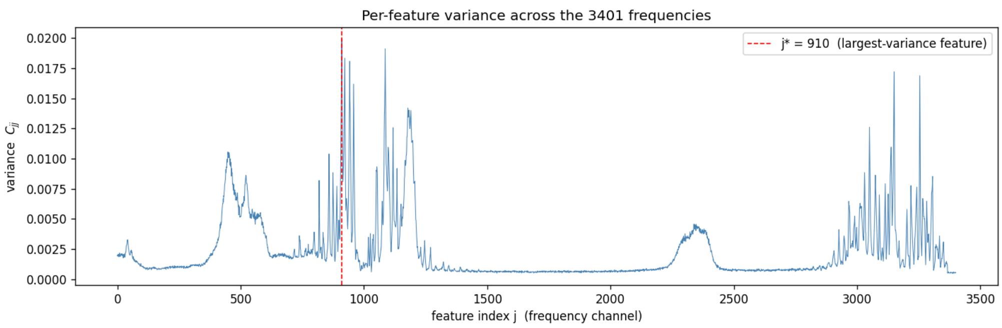

Figure 2: Per-feature variance C_{jj}; the maximum is at $j^\star = 910$ (red).

4 Part (c): the eigenspectrum (PCA)

Proposition 3. *For unit w, the variance of the data along w is $w^\top C w$; if $C\nu = \lambda\nu$ with $\|\nu\| = 1$ then $\nu^\top C \nu = \lambda$. Moreover C is positive semi-definite, so every $\lambda_i \geq 0$.*

Proof. The projections $z_i = \tilde{x}_i \cdot w$ have variance $\frac{1}{N}\sum_i z_i^2 = w^\top C w$; substituting an eigenvector gives λ. And $w^\top C w = \frac{1}{N}\|\widetilde{X} w\|^2 \geq 0$, so $\lambda = \nu^\top C \nu \geq 0$. $\square$

By the spectral theorem C has real eigenvalues and an orthonormal eigenbasis; we use `numpy.linalg.eigh` (symmetric solver: real eigenvalues, orthonormal eigenvectors, faster and stabler than `eig`), sort decreasingly, and clip rounding-level negatives to 0.

The dominance ratio is $\lambda_1/\lambda_2 \approx 3.53$: one direction dominates, indicating strong low-dimensional structure (a wide spectral gap, the condition for fast Power-Method convergence, Question 2).

k	1	2	3	4	5	6	7	8	9
λ_k	2.492	0.705	0.590	0.254	0.123	0.105	0.078	0.039	0.0093

Table 1: Leading eigenvalues. The drop $\lambda_8/\lambda_9 \approx 4.1$ is the largest in the spectrum.

The *elbow* is the largest multiplicative drop between consecutive eigenvalues (a ratio, since the signal/noise boundary is multiplicative on the log scale of Figure 3): $\lambda_8/\lambda_9 \approx 4.1$ gives $k^\star = 8$. Here λ_8 remains $\approx 4\times$ above the noise floor (signal) while λ_9 lies on it (noise); this matches the ~ 10 dominant molecules. The eight signal components explain only $\approx 55\%$ of the variance, and reaching 95% needs $k = 667$ components (Table 2), because the noise floor is spread over ≈ 900 dimensions; the "95%" rule therefore overstates the dimensionality, and the elbow is the honest measure. Check: $\sum_i \lambda_i \approx \operatorname{tr} C \approx 7.90$ (Prop. 2).

cumulative variance	50%	80%	90%	95%	99%
components k	4	309	516	667	850

Table 2: Components to reach each threshold: contrast the 8 signal components (55%) with 667 for 95%.

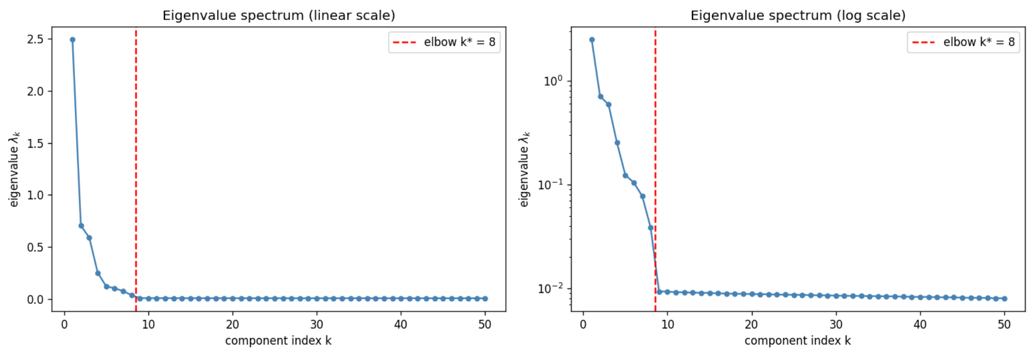

Figure 3: Eigenvalue spectrum (linear and log). The log plot shows a sharp elbow at $k^\star = 8$, then a flat noise floor.

5 Part (d): the leading eigenvector

$\nu_1 \in \mathbb{R}^M$ (feature space) gives each frequency a *loading*; the *scores* $z_i = \tilde{x}_i \cdot \nu_1$ are the maximally-spread one-dimensional summaries (Prop. 3). Eigenvectors are sign-ambiguous; we fix the largest loading positive. The loadings are supported on two narrow bands ($j \approx 900$–960, 3150–3260) with $\approx 27\%$ negative entries, so ν_1 is a *contrast* (balance of absorption between bands), not overall intensity. With cosine similarity $\cos(a, b) = \frac{a \cdot b}{\|a\|\|b\|}$ (scale-invariant), ν_1 matches a *centered* spectrum at $\cos \approx 0.92$, but only 0.67 against the raw spectrum and 0.03 against the mean: living in the centered space, ν_1 is $\approx$orthogonal to the mean and is a *difference-spectrum* (Figure 4), echoing Question 1's dominant-eigenvector steady state.

6 Part (e): clustering in covariance space

With $V_4 = [\nu_1\, \nu_2\, \nu_3\, \nu_4]$, the compressed coordinates $Z = \widetilde{X} V_4 \in \mathbb{R}^{\widetilde{N} \times 4}$ give each vial four scores; the $\binom{4}{2} = 6$ pairwise scatters (Figure 5) show many tight, separated groups. Counting by leader

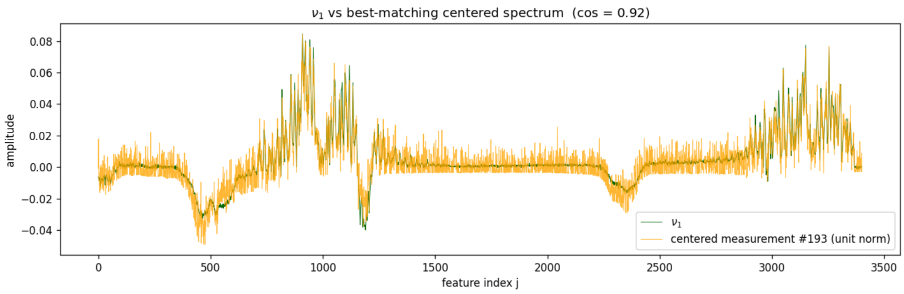

Figure 4: ν_1 (green) overlaid with the best-matching centered spectrum (cos ≈ 0.92); a difference-spectrum on two molecular bands.

clustering (a numpy-only threshold method) gives 25 clusters, stable for radii in $[6, 12]\times$ the median nearest-neighbour spacing — a property of the data, not the threshold. Thus the 924 vials form ≈ 25 concoctions, below the "fewer than 100" of Part (h). Although $\nu_1, \dots, \nu_4$ live in feature space, projecting samples onto them exposes sample-space structure, since similar spectra get similar scores. Part (f) repeats this with the Gram matrix $G = \frac{1}{\widetilde{N}}\widetilde{X}\widetilde{X}^{\mathsf{T}}$ (same nonzero eigenvalues as C, Question 2d), which should recover the same groups.

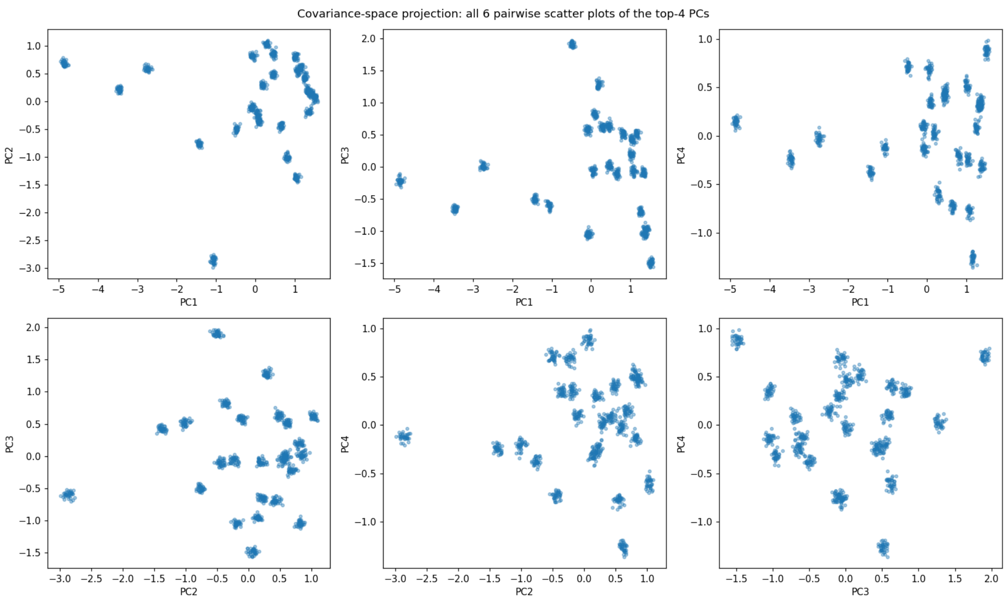

Figure 5: The six pairwise scatters of the top-4 PC scores; each point is one vial. The data resolves into ≈ 25 clusters.